

Erdős Problem #836

EP-836

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $r \geq 2$ and G be a r -uniform hypergraph with chromatic number 3 (that is, there is a 3-colouring of the vertices of G such that no edge is monochromatic).

Suppose any two edges of G have a non-empty intersection. Must G contain $O(r^2)$ many vertices? Must there be two edges which meet in $\gg r$ many vertices?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>